Periodic Functions and Their Transforms

Contents

Part I — Condensed Cheat Sheet

3.6 Periodic Functions and Their Transforms

Unit at a Glance: Core Sub-Topics

- **Motivation:** Modeling repeating physical phenomena like AC circuits, structural vibrations, and signal processing.
- **Master Formula:** Combining integration over a single period with the geometric series multiplier.
- **Constructive Tools:** Using Heaviside step functions $u_c(t)$ and Dirac Delta functions $\delta(t-c)$ to build piecewise and instantaneous periodic signals.

Origin & Motivation

In engineering and physics, systems rarely experience isolated, one-off forces. Instead, they are driven by repeating cyclic inputs—an alternating current pushing through a circuit, an engine vibrating at a constant RPM, or a digital clock generating continuous square waves.

To solve Ordinary Differential Equations modeling these systems, we need a way to take the Laplace transform of a function that stretches endlessly into the future. Integrating an infinite sequence of repeating intervals is intimidating, but by leveraging the repeating nature of the function, we can compress the infinite work down to evaluating just one single cycle.

Intuition & Derivation

Let $f(t)$ be a periodic function with period P, meaning $f(t+P) = f(t)$ for all $t \geq 0$. The Laplace transform is defined as:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$$

We can break this infinite integral into a sum of integrals over each period P:

$$\int_0^P e^{-st} f(t)\,dt + \int_P^{2P} e^{-st} f(t)\,dt + \int_{2P}^{3P} e^{-st} f(t)\,dt + \dots$$

The key move is a *time-shift substitution*. Consider the general n-th term, integrated over the interval $[nP, (n+1)P]$. We substitute $t = \tau + nP$, so that $dt = d\tau$. As t runs across $[nP, (n+1)P]$, the new variable τ runs across $[0, P]$, collapsing every interval back onto the very first period:

$$\int_{nP}^{(n+1)P} e^{-st} f(t)\,dt = \int_0^P e^{-s(\tau+nP)} f(\tau + nP)\,d\tau$$

Because f has period P, repeating the shift n times leaves the function unchanged: $f(\tau + nP) = f(\tau)$. Meanwhile, the exponential splits by the law of exponents:

$$e^{-s(\tau+nP)} = e^{-s\tau} \cdot e^{-nPs}$$

The factor e^{-nPs} is constant with respect to τ, so it pulls straight out of the integral:

$$\int_{nP}^{(n+1)P} e^{-st} f(t)\,dt = e^{-nPs} \int_0^P e^{-s\tau} f(\tau)\,d\tau$$

Summing this identity over every cycle $n = 0, 1, 2, \dots$ factors the shared single-period integral out front, leaving behind a sum of the constants e^{-nPs}:

$$\left(1 + e^{-Ps} + e^{-2Ps} + e^{-3Ps} + \dots\right) \int_0^P e^{-s\tau} f(\tau)\,d\tau$$

The expression in the parentheses is a classic infinite geometric series with ratio $r = e^{-Ps}$. Since $e^{-Ps} < 1$ for $s > 0$, the series converges to $\frac{1}{1-e^{-Ps}}$.

Periodic Function Transform

If $f(t)$ is piecewise continuous for $t \geq 0$ and has period P, its Laplace transform is:

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-Ps}} \int_0^P e^{-st} f(t)\,dt$$

Algorithm: Transforming a Periodic Function

1. **Identify the Period (P):** Determine the exact length of one complete cycle.
2. **Define the Single-Cycle Function:** Write an explicit piecewise or Heaviside expression for $f(t)$ strictly on the interval $[0, P)$.
3. **Integrate over $[0, P]$:** Evaluate $\int_0^P e^{-st} f(t)\,dt$.
4. **Apply the Multiplier:** Multiply your integral result by $\frac{1}{1-e^{-Ps}}$.
5. **Simplify Algebraically:** Look for factoring opportunities, especially difference of squares like $(1 - e^{-2s})(1 + e^{-2s})$.

Common Pitfall: Integration Bounds

Do not integrate to infinity! A highly common mistake is forgetting to bound the integral $\int_0^P e^{-st} f(t)\, dt$ exclusively to the first period. The $\frac{1}{1-e^{-Ps}}$ term already accounts for all future cycles $[P, \infty)$.

Advanced Machinery

To build complex waveforms, we utilize specific mathematical tools:

- **Heaviside Step Function $u_c(t)$:** Used to "turn on" or "turn off" pieces of a function within the period $[0, P]$.
- **Dirac Delta Function $\delta(t-c)$:** Used to represent sudden, instantaneous strikes or impulses within a repeating cycle.

Worked Example 1: Square Wave

For $f(t) = +1$ on $[0, 2)$ and -1 on $[2, 4)$ with period $P = 4$:
First, we handle the integral breakdown over the period $[0, 4]$:

$$\int_0^4 e^{-st} f(t)\, dt = \int_0^2 (1)e^{-st}\, dt + \int_2^4 (-1)e^{-st}\, dt$$

Evaluating the integrals:

$$\left[\frac{-1}{s}e^{-st}\right]_0^2 - \left[\frac{-1}{s}e^{-st}\right]_2^4 = \left(\frac{1-e^{-2s}}{s}\right) - \left(\frac{e^{-2s}-e^{-4s}}{s}\right) = \frac{1-2e^{-2s}+e^{-4s}}{s} = \frac{(1-e^{-2s})^2}{s}$$

Next, we address the geometric multiplier denominator $1 - e^{-Ps} = 1 - e^{-4s}$. Using difference of squares for algebraic simplification:

$$1 - e^{-4s} = (1 - e^{-2s})(1 + e^{-2s})$$

Bringing it all together in the master formula:

$$F(s) = \frac{1}{1-e^{-4s}} \int_0^4 e^{-st} f(t)\, dt = \frac{(1-e^{-2s})^2}{s(1-e^{-2s})(1+e^{-2s})} = \frac{1-e^{-2s}}{s(1+e^{-2s})}.$$

Worked Example 2: Impulse Train

Find the transform of a periodic unit impulse train $f(t) = \sum_{n=0}^{\infty} \delta(t - nP)$.
Here, the period is P. Over a single period $[0, P]$, the function is simply $f(t) = \delta(t)$. Integrating over the first period yields:

$$\int_0^P e^{-st}\delta(t)\,dt = e^{-s(0)} = 1$$

Applying the master formula:

$$F(s) = \frac{1}{1 - e^{-Ps}} \int_0^P e^{-st} f(t)\,dt = \frac{1}{1 - e^{-Ps}}(1) = \frac{1}{1 - e^{-Ps}}.$$

Unit Key Takeaway

The periodic Laplace transform converts an infinite sequence of repeating integrations into a single finite integral multiplied by the geometric sum factor $\frac{1}{1-e^{-Ps}}$.

Part II — Practice Set

Problem 1: The Sawtooth Wave
Find the Laplace transform of the periodic function $f(t) = t$ for $0 \leq t < 2$, with period $P = 2$.

Problem 2: Rectified Sine Wave
Find the Laplace transform of the fully rectified sine wave $f(t) = |\sin(t)|$. *Hint: The absolute value cuts the normal sine period in half. What is the new period P?*

Problem 3: Heaviside Formulation
Find the Laplace transform of the periodic function defined over its first period $P = 3$ by:

$$f(t) = u_1(t) - u_2(t) \quad \text{for} \quad 0 \leq t < 3$$

Part III — Complete Solutions

Solution 1: The Sawtooth Wave
The period is $P = 2$. Applying the master formula:

$$F(s) = \frac{1}{1-e^{-2s}} \int_0^2 e^{-st} t \, dt$$

We use Integration by Parts for $\int_0^2 te^{-st}\,dt$ with the choices $u = t$, $du = dt$, $dv = e^{-st}\,dt$, $v = -\frac{1}{s}e^{-st}$:

$$\int_0^2 te^{-st}\,dt = \left[-\frac{t}{s}e^{-st}\right]_0^2 - \int_0^2 \left(-\frac{1}{s}e^{-st}\right) dt$$

$$= \left(-\frac{2}{s}e^{-2s} - 0\right) - \left[\frac{1}{s^2}e^{-st}\right]_0^2$$

$$= -\frac{2}{s}e^{-2s} - \left(\frac{1}{s^2}e^{-2s} - \frac{1}{s^2}\right) = \frac{1}{s^2} - \frac{1}{s^2}e^{-2s} - \frac{2}{s}e^{-2s}$$

Factoring out the terms and combining with the multiplier:

$$F(s) = \frac{1}{1-e^{-2s}} \left(\frac{1-e^{-2s}-2se^{-2s}}{s^2}\right)$$

Solution 2: Rectified Sine Wave
For $f(t) = |\sin(t)|$, the normal period of 2π is halved to $P = \pi$. Over the interval $[0, \pi)$, $\sin(t)$ is positive, so $f(t) = \sin(t)$.

$$F(s) = \frac{1}{1-e^{-\pi s}} \int_0^\pi e^{-st} \sin(t)\,dt$$

The standard integral identity $\int e^{at}\sin(bt)\,dt = \frac{e^{at}}{a^2+b^2}(a\sin(bt) - b\cos(bt))$ gives us:

$$\int_0^\pi e^{-st}\sin(t)\,dt = \left[\frac{e^{-st}}{s^2+1}(-s\sin(t) - \cos(t))\right]_0^\pi$$

Evaluating at the bounds ($\sin(\pi) = 0$, $\cos(\pi) = -1$ and $\sin(0) = 0$, $\cos(0) = 1$):

$$= \left(\frac{e^{-\pi s}}{s^2+1}(0-(-1))\right) - \left(\frac{1}{s^2+1}(0-1)\right) = \frac{e^{-\pi s}+1}{s^2+1}$$

Multiplying by the periodic factor gives the combined transform:

$$F(s) = \frac{e^{-\pi s}+1}{(1-e^{-\pi s})(s^2+1)}$$

To collapse this into the standard hyperbolic form, factor $e^{-\pi s/2}$ out of the numerator and denominator of the periodic ratio:

$$\frac{1+e^{-\pi s}}{1-e^{-\pi s}} = \frac{e^{\pi s/2}+e^{-\pi s/2}}{e^{\pi s/2}-e^{-\pi s/2}} = \frac{2\cosh(\pi s/2)}{2\sinh(\pi s/2)} = \coth\left(\frac{\pi s}{2}\right)$$

Therefore the rectified sine transform reduces to the compact closed form:

$$F(s) = \frac{1}{s^2 + 1} \coth\left(\frac{\pi s}{2}\right)$$

Solution 3: Heaviside Formulation

The period is $P = 3$. The function is 1 between $t = 1$ and $t = 2$, and 0 elsewhere in the period.

$$F(s) = \frac{1}{1 - e^{-3s}} \int_0^3 e^{-st}(u_1(t) - u_2(t))\, dt$$

This simplifies the integral bounds to $[1, 2]$:

$$\int_1^2 e^{-st}(1)\, dt = \left[-\frac{1}{s}e^{-st}\right]_1^2 = -\frac{1}{s}(e^{-2s} - e^{-s}) = \frac{e^{-s} - e^{-2s}}{s}$$

Combining with the multiplier:

$$F(s) = \frac{e^{-s} - e^{-2s}}{s(1 - e^{-3s})}$$